

Hand in

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Problem (1). Let $X = [-1, 1] \times \{-1, +1\} \subseteq \mathbb{R}^2$ have the usual metric topology and define a relation $\sim$ on X saying that $(a, -1) \sim (a, 1)$ for all $a \in [-1, 0) \cup (0, 1]$. Let $Y = X / \sim$ have the quotient topology

- Check that $\sim$ is an equivalence relation (nevermind I guess?)
- Is Y Hausdorff?
- Does Y have a countable basis for its topology?
- Check up the definition of connectedness in section 23: is Y connected?
- Check up the definition of compact in section 26: is Y compact?

Solution and proof of all.

Hausdorff?

Recall that Y is Hausdorff if any two distinct points have some open neighborhoods that are disjoint.

We will use the fact that open sets in X correspond exactly with open sets in Y .

For Y to have the Hausdorff property we would require that, as $(0, 1)$ and $(0, -1)$ are distinct, they have some open disjoint neighborhoods that are disjoint.

Denote $(a, 1) = a_+$ and $(0, -1) = a_-$. Let U be an open neighborhood of 0_+ and V an open neighborhood of 0_- .

Then $q^{-1}(U), q^{-1}(V)$ are open neighborhoods of X . As X has the metric topology we know there are $\varepsilon_+, \varepsilon_- > 0$ such that

$$(-\varepsilon_+, \varepsilon_+) \times \{1\} \subseteq q^{-1}(U)$$

and

$$(-\varepsilon_-, \varepsilon_-) \times \{-1\} \subseteq q^{-1}(V).$$

Choose $0 < a < \min\{\varepsilon_-, \varepsilon_+\}$. Then $a_+ \in q^{-1}(U)$ and $a_- \in q^{-1}(V)$, but

$$a_+ \sim a_-$$

so

$$q(a_+) = q(a_-) \in Y.$$

Call this point y . Then $y \in U \cap V$, meaning they are not disjoint. This violates the Hausdorff property. Intuitively, we will use the fact that some open neighborhood of $(0, 1)$ will have elements whose fibers

Countable basis?

Indeed it does have a countable basis.

X has a countable basis since it is just two copies of $[-1,1] \subseteq \mathbb{R}$ which has a countable basis which we can construct using the rationals.

Let $\mathcal{B} = \{B_1, B_2, \dots\}$ be such a basis. Since q is open ($q^{-1}q(B)$ is open),

$$\mathcal{C} = \{q(B) \mid B \in \mathcal{B}\}$$

consists of open sets in Y . Moreover, if $y \in U$ for some open set in Y , choose $x \in q^{-1}(U)$. Since $q^{-1}(U)$ is open, there is some $B \in \mathcal{B}$ with

$$x \in B \subseteq q^{-1}(U).$$

Therefore $Y \in q(B) \subseteq U$. So $\mathcal{C}$ is a countable basis for Y .

Connected?

The space is indeed connected. I like to use the following procedure.

Let $f : Y \rightarrow \{0,1\}$ be any continuous function where the codomain has the discrete topology.

Let $X_+ = [-1,1] \times \{1\}$ and $X_- = [-1,1] \times \{-1\}$. $f \circ q$ is continuous so the restrictions

$$f \circ q \big|_{X_+} : X_+ \rightarrow \{0,1\}$$

and

$$f \circ q \big|_{X_-} : X_- \rightarrow \{0,1\}$$

are continuous. X_+ and X_- are homeomorphic to $[-1,1]$ and therefore connected so these maps must be constant. For convenience write

$$f \circ q \big|_{X_+} \equiv c_+, \text{ and } f \circ q \big|_{X_-} \equiv c_-$$

where $c_+, c_- \in \{0,1\}$.

Recall that $X_+ \ni a_+ \sim a_- \in X_-$ so $q(a_+) = q(a_-)$. Hence

$$c_+ = f(q(a_+)) = f(q(a_-)) = c_-.$$

As $Y = q(X_+) \cup q(X_-)$ the map f must be constant on the whole domain.

Hence the space is connected!

Compact?

The space is indeed compact.

We know that $[-1,1] \times \{1\}$, $[-1,1] \times \{-1\} \cong [-1,1]$ are compact. So for a covering of X reduce the open covers on each copy of the line to a finite cover.

Now take some open cover $\bigcup_{\alpha} U_{\alpha}$ of Y . Then

$$q^{-1} \left(\bigcup_{\alpha} U_{\alpha} \right) = \bigcup_{\alpha} q^{-1}(U_{\alpha})$$

is a cover of X . Since X is compact we can take a finite number of these

$$\bigcup_{\alpha_i, i \in [n]} q^{-1}(U_\alpha)$$

to cover X . Meaning, every a_+ is contained in some U_{α_i} and every a_- is contained in some U_{α_j} . Then $q(a_+), q(a_-) \in U_{\alpha_i} \cup U_{\alpha_j}$ so the image of this finite cover suffices as a finite covering of Y .

It would also have sufficed to point out that the continuous image of a compact space is compact! $\square$

Problem (2). Let $\mathbb{R}^\omega$ be the countably infinite product of $\mathbb{R}$ with itself. Let $\mathbb{R}^\infty \subset \mathbb{R}^\omega$ be the subset of all sequences $(x_1, x_2, \dots)$ that are eventually zero (i.e., there exists an N such that $x_n = 0$ for all $n > N$).

- a) What is the interior and closure of $\mathbb{R}^\infty$ in $\mathbb{R}^\omega$ if $\mathbb{R}^\omega$ is given the product topology? Is $\mathbb{R}^\infty$ (with this topology) Hausdorff?
- b) What is the interior and closure of $\mathbb{R}^\infty$ in $\mathbb{R}^\omega$ if $\mathbb{R}^\omega$ is given the box topology? Is $\mathbb{R}^\infty$ (with this topology) Hausdorff?
- c) What is the interior and closure of $\mathbb{R}^\infty$ in $\mathbb{R}^\omega$ if $\mathbb{R}^\omega$ is given the uniform topology? Is $\mathbb{R}^\infty$ (with this topology) Hausdorff?

Solution and proof to (a). $\square$